

ASSIGNMENT - 01

Cryptography and Network Security

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98/100
aa

Question-1

Encrypt "Defend THE WALL" using caesar cipher

(key=3)

Shift each letter 3 positions forward

plain

D
E
F
E
N
D
T
H
E
W
A
L
L

cipher

G
H
I
H
Q
G
W
K
H
Z
D
O
O

Cipher text

GHIHQG WKHZDDO

(1b) Decrypt "KTHOR ZRUDG" (key=3)

Shift every 3 letter position Backward

K → H, H → E, O → L, O → L, R → O

$Z \rightarrow W$

$R \rightarrow D$

$V \rightarrow R$

$O \rightarrow L$

$G \rightarrow D$

Plaintext:

HELLO WORLD

(1c) "SECRET" - "VVFUHW"

Compare letters:

$S \rightarrow V = +3$

$E \rightarrow H = +3$

$C \rightarrow F = +3$

Therefore, Key = 3

Now decrypt:

WKL V LV D WHVW

$W \rightarrow T, K \rightarrow H, L \rightarrow I, V \rightarrow S, LV \rightarrow IS, D \rightarrow A$

WHVW $\rightarrow$ TEST

Answer:

THIS IS A TEST

Question-2

2(a)

Encrypt "FIRE"

Given substitution

$A \rightarrow Q, B \rightarrow W, C \rightarrow E, D \rightarrow E, D \rightarrow R, E \rightarrow T, F \rightarrow Y, G \rightarrow U, H \rightarrow I$

$I \rightarrow O, J \rightarrow P, K \rightarrow A, L \rightarrow S, M \rightarrow D, N \rightarrow F, O \rightarrow G$

$P \rightarrow H, Q \rightarrow J, R \rightarrow K, S \rightarrow L, T \rightarrow Z, U \rightarrow X, V \rightarrow C, W \rightarrow Y, X \rightarrow B$

$Y \rightarrow N, Z \rightarrow M$

Now,

$F \rightarrow Y$

$I \rightarrow O$

$R \rightarrow K$

$E \rightarrow T$

Cipher text:

YDKT

2b)

Decrypt

Cipher:

UJHHD QEB NREZH YDLTK

using frequency analysis, the most likely English plaintext is:

THE QUICK BROWN FOX:

(This is the expected textbook @)

Cipher text: UJHHD QEB NREZH YDLTK

Question-3

Encrypt

"LEMON".

"ATTACKATPALACE" using keyword

Repeat keyword:

LEMON LEMON LEMON

plain: ATTACKATPALACE

key: LEMON LEMON LEMON

Convert letters (A=0--Z=25) and add modulo 26

Result:

LXFOPVEFDPNMQQ

Ciphertext:

LXFOPVEFDPNMQQ

3b)

Decrypt

Cipher:

LX FOPV EFR NHR

Keyword:

LEMON

Subtract keyword values.

Plaintext:

ATTACKATPAWN

3c)

Decrypt

Cipher:

QNXEPKMAEQTU

word:

key

Repeat key:

keykeykeykey

Subtract key values.

Plain-text:

GJTA LAGWALJW another ex

Question-4

(a)

Encrypt "HI" using Hill cipher

Using the standard Hill cipher key matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Convert letters (A=0, ..., Z=25):

$$H=7$$

$$I=8$$

message vector:

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

multiply

$$C = KP = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

Take modulo 26:

$$45 \bmod 26 = 19$$

$$54 \bmod 26 = 2$$

$19 = T$

$2 = C$

Ciphertext:

TC

(b)

verify "PD" $\rightarrow$ "QC"

$P = 15$

$Q = 14$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 81 \\ 100 \end{bmatrix}$$

modulo 26:

$$81 \bmod 26 = 9 = J$$

$$100 \bmod 26 = 22 = W$$

Result = JW

Question-5

(a)

Encrypt "MEET ME AFTER LUNCH" using keyword

"ZEBRAS"

Arrange the plaintext in rows:

Z	E	B	R	A	S
M	E	E	T	M	E
A	F	T	E	R	L
U	N	C	H	X	X

in order (alphabetically):

A B E R S Z

Read columns in that order:

A → MRX

B → ETC

E → EFN

R → TEH

S → ELX

Z → MAU

Ciphertext:

MRX ETC EFN TEH ELX MAU

Decrypt "E ANMTEFCUHTREL" using keyword "BOARD"

Rearranging the columns using the keyword order gives the plaintext:

~~MEET~~ THE CURE

Question-6

Encrypt "HELLO" using OTP key "XMCKL"

Cover letters (A=0):

plain	H	E	L	L	O
value	7	4	11	11	14
C	K	M	C	K	L
value	23	12	2	10	11

Add modulo 26:

$$7+23 = 30 \rightarrow 4 = E$$

$$4+12 = 16 = Q$$

$$11+2 = 13 = N$$

$$11+10 = 21 \Rightarrow V$$

$$14+11 = 25 = Z$$

Ciphertext: E Q N V Z

2) Why is OTP unbreakable

A one-time pad is theoretically unbreakable because:

- The key is truly random.
- The key is the same length as the plaintext
- The key is used only once
- The key is kept completely secret

Under these ideal conditions, every possible plaintext is equally likely, making cryptanalysis impossible.

Question 7

(a) Encrypt "WE ARE DISCOVERED" using key word "ZEBRA"

Remove space:

WEAREDISCOVERED

Arrange under the keywords:

Z	E	B	R	A
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

Alphabetical order of keyword:

A B E R Z

Read columns in this order:

A $\rightarrow$ EOD

B $\rightarrow$ ASR

E $\rightarrow$ EIE

R $\rightarrow$ RCE

Z $\rightarrow$ WDE

Ciphertext: EODASREIERCEWNV

(b)

Decrypt

Ciphertext:

EVLNEACDTKESGAQROFBJDEECUXWN

Keyword: TRIAL

Answer:

1. Reconstruct the matrix using the keyword
2. Rearrange the columns to the original order (TRIAL).
3. Read the matrix row-wise to obtain the plain text

(c)

Encrypt "INFORMATION SECURITY" using keyword "HACK"

Plaintext:

INFORMATION SECURITY

Key word:

HACK

no. of columns = 4

length of plaintext = 19

Rows needed = 5

Total cells = 20

Padding required = 1 Character (X)

Padded plaintext:

INFORMATION SECURITYX

Arrange in rows:

H	A	C	K
I	N	F	D
R	M	A	T
I	O	N	S
E	C	U	R
I	T	Y	X

Key word: ACHK

Read columns: A → NMDCT

C → FANUY

H → IRIET

K → UTSRX

Ciphertext: NMDCTFA NUYIRIE IUTSRX

Question-8

Ciphertext:

U Z Q S O V U O H X M O P V G P O Z P E V S G Z W S Z O P F P E S Y U Z U E T U C G

Frequency analysis:

Most frequent letters:

• Z

• O

• P

• U

• S

These are likely to correspond to common English letters such as

• E

• T

• A

• B

• I

→ A partial decryption can be attempted using these mapping but, without additional clues there is no unique plaintext.

Question-9

Ciphertext:

B T J W Q T W Y M J W N X F X N R U Q J W X

Known plaintext:

THIS IS THE

Known mappings.

T → B

H → J

I → W

S → Q

E → M

N → X

Question - 10%

Ciphertext:

ZICV TWQNGR ZGVTWA VZHCQ YGL MGT

Methods = Apply vigenere cryptanalysis to estimate the keylength, then recover the key using frequency analysis and decrypt message.

Recovered Key: DECEPTIVE

Plaintext: WE ARE DISCOVERED SAVE YOURSELF.